Random Password Generator

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Overview of Password Security

A random password generator is essential for creating **secure passwords** using various character types. This project emphasizes the importance of strong passwords in protecting sensitive data against cyber threats.

The project aims to **understand random password generation** in C++, emphasizing string manipulation to create secure passwords dynamically. Users can set desired lengths and interact via a simple menu-driven interface.

Objectives and Features of the Project

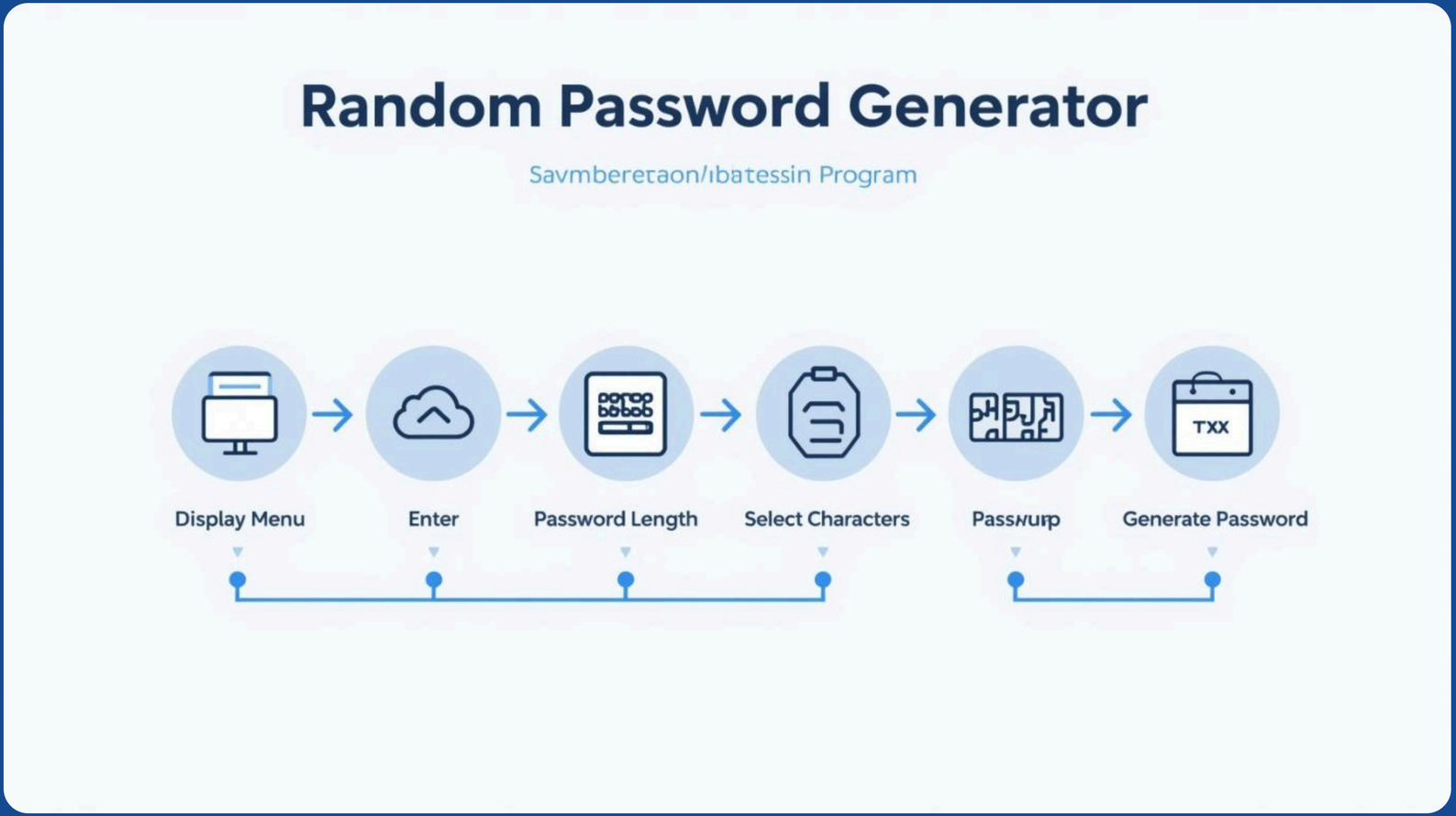
Objectives:

- Understand random number generation in C++
- Learn string manipulation
- Create secure, user-defined passwords

Features:

- User inputs desired password length
- Random password generation
- Includes letters, numbers, and symbols

Program Logic and Workflow



Working Steps:

1. Display menu (Generate Password / Exit)
2. User enters password length
3. Program selects random characters
4. Password is generated and displayed

Loops Used:

- while loop for menu control
- for loop for character generation

Design: Flowchart-style layout with code and loop icons

```
----- Random Password Generator -----
1. Generate Password
2. Exit
Enter your choice: 1
Enter password length: 8
Generated Password: Px9$RYDi
```

Conclusion

Example Output:

Generated Password: Px9$RYDi

Conclusion:

The project demonstrates the use of random number generation, loops, and string manipulation in C++ to create secure passwords.

It highlights practical programming concepts essential for cybersecurity and software development.